In the above display $\nu = \nu_+ - \nu_-$ stands for a Hahn-Jordan decomposition of the measure, while $\mathrm{osc}_\varphi(f)$ stands for the φ-oscillation of a function $f \in \mathcal{B}(\mathbb{X})$, which is given by

$$\mathrm{osc}_\varphi(f) := \sup_{x_1,x_2} \frac{|f(x_1) - f(x_2)|}{\varphi(x_1) + \varphi(x_2)} \leqslant \|f\|_\varphi.$$

For a detailed proof of the equivalent formulations in the latter definition (3.2), we refer to Proposition 8.2.16 in [44]. The weighted Wasserstein formulation (1.7) of the φ-norm is a consequence of Kellerer's extension of the Kantorovich-Rubinstein theorem [55, 71, 70], see also the recent article [39].

The choice of condition $\varphi \geqslant 1/2$ in the definition of the φ-norm given in (3.2) is imposed only to recover the conventional total variation dual distance $|||\mu_1 - \mu_2|||_{1/2} = \|\mu_1 - \mu_2\|_{\mathsf{tv}}$ between probability measures $\mu_1, \mu_2 \in \mathcal{M}_1(\mathbb{X}) = \mathcal{M}_{1,\frac{1}{2}}(\mathbb{X})$, where $\mathcal{M}_{1,\varphi}(\mathbb{X})$ denotes the subset of probability measures ν s.t. $\nu(\varphi) < \infty$. Recall that, given an arbitrary Polish space S, for any $\mu_1, \mu_2 \in \mathcal{M}_1(S)$ and $\epsilon \in]0,1]$ we have

$$\|\mu_1 - \mu_2\|_{\mathsf{tv}} \leqslant 1 - \epsilon \quad \text{if, and only if,} \quad \exists \nu \in \mathcal{M}_1(S) \ : \ \mu_1 \geqslant \epsilon\nu \quad \text{and} \quad \mu_2 \geqslant \epsilon\nu. \tag{3.3}$$

Choosing $\varphi \geqslant 1$ and $\kappa > 0$, for any $\nu \in \mathcal{M}_b(\mathbb{X})$ we also have the norm equivalence formulae

$$\begin{aligned} &\varphi_\kappa := 1 + \kappa\varphi \\ &\Longrightarrow \kappa\ |||\nu|||_\varphi \leqslant |||\nu|||_{\varphi_\kappa} \leqslant (1+\kappa)|||\nu|||_\varphi \quad \text{and} \quad |||\nu|||_{\varphi_\kappa} \geqslant |||\nu|||_1 \geqslant \|\nu\|_{\mathsf{tv}}. \end{aligned} \tag{3.4}$$

For exponential functions of the form

$$\varphi(x) = c\ e^{g(x)} \text{ with some parameter } c \geqslant 0 \text{ and } g(x_1) + g(x_2) \geqslant \varrho_g(x_1, x_2)$$

for some semi-metric ϱ_g we also have the estimate

$$\varphi(x_1) + \varphi(x_2) \geqslant 2c\ e^{g(x_1)/2} e^{g(x_2)/2} \geqslant \varrho_\varphi(x_1, x_2) := 2c\ \left(e^{\frac{1}{2}\varrho_g(x_1,x_2)} - 1\right).$$

Also note that for any $p \geqslant 1$ we have

$$\varrho_\varphi(x_1, x_2) \geqslant c_p\ \varrho_g(x_1, x_2)^p \quad \text{with} \quad c_p := \frac{c}{2^{p-1}p!}$$

and this implies that

$$\frac{c}{2^{p-1}p!}\ \mathbb{W}_{p,\varrho_g}(\mu_1, \mu_2)^p \leqslant \mathbb{W}_{\varrho_\varphi}(\mu_1, \mu_2) \leqslant |||\mu_1 - \mu_2|||_\varphi. \tag{3.5}$$

Last but not least, on normed vector spaces consider a function of the form

$$\varphi(x) \geqslant a + \|b(x)\|^p$$

for some function $b(x)$ and $a, p > 0$. In this case we have

$$\varphi(x_1) + \varphi(x_2) \geqslant \varrho(x_1, x_2) \quad \text{with} \quad \varrho(x_1, x_2) := \frac{1}{2^{(p-1)_+}}\ \|b(x_1) - b(x_2)\|^p \tag{3.6}$$

as well as the Wasserstein estimate (1.10).

3.2. **Some contraction coefficients.** Let δ_x denote the Dirac delta measure located at x. Whenever $\delta_{x_1}\mathcal{K} \ll \delta_{x_2}\mathcal{K}$ for any $(x_1, x_2) \in \mathbb{X}^2$, we denote by $\imath_{x_1,x_2}(\mathcal{K})(y)$ the Radon-Nikodym function

$$\imath_{x_1,x_2}(\mathcal{K})(y) := \frac{d\delta_{x_1}\mathcal{K}}{d\delta_{x_2}\mathcal{K}}(y). \tag{3.7}$$

We also consider the $[0,1]$-valued parameters

$$\begin{aligned} \jmath(\mathcal{K}) &:= \inf_{(x_1,x_2)\in\mathbb{X}^2} \inf_{y\in\mathbb{Y}} \imath_{x_1,x_2}(\mathcal{K})(y) \quad \text{and} \\ \hbar(\mathcal{K}) &:= \inf_{(x_1,x_2)\in\mathbb{X}^2} \inf_{(y_1,y_2)\in\mathbb{Y}^2} (\imath_{x_1,x_2}(\mathcal{K})(y_1)\ \imath_{x_2,x_1}(\mathcal{K})(y_2)) \geqslant \jmath(\mathcal{K})^2, \end{aligned} \tag{3.8}$$

and we note that

$$\delta_{x_1}\mathcal{K}(dy_1)\ \delta_{x_2}\mathcal{K}(dy_2) \ \geqslant \hbar(\mathcal{K})\ \delta_{x_2}\mathcal{K}(dy_1)\ \delta_{x_1}\mathcal{K}(dy_2).$$

Integrating the above inequality w.r.t. the variable y_2 we readily see that

$$\delta_{x_1}\mathcal{K}(dy_1) \ \geqslant \hbar(\mathcal{K})\ \delta_{x_2}\mathcal{K}(dy_1), \quad \text{which readily yields} \quad \jmath(\mathcal{K}) \geqslant \hbar(\mathcal{K}) \geqslant \jmath(\mathcal{K})^2. \tag{3.9}$$

3.2.1. *Birkhoff contraction coefficient.* The Birkhoff contraction coefficient of a Markov transition $\mathcal{K}$ from $\mathbb{X}$ into $\mathbb{Y}$ is defined by

$$\chi_H(\mathcal{K}) := \sup \frac{H(\nu_1\mathcal{K},\nu_2\mathcal{K})}{H(\nu_1,\nu_2)} = \frac{1-\sqrt{\hbar(\mathcal{K})}}{1+\sqrt{\hbar(\mathcal{K})}}, \tag{3.10}$$

where the supremum is taken over all comparable probability measures $\nu_1,\nu_2 \in \mathcal{M}_1(\mathbb{X})$ such that $H(\nu_1,\nu_2)>0$ and the second equality is explicitly derived in Appendix B. Combining (3.9) and (3.10) we readily obtain the equivalent inequalities

$$\chi_H(\mathcal{K}) < 1 \Longleftrightarrow \hbar(\mathcal{K}) > 0 \Longleftrightarrow \jmath(\mathcal{K}) > 0.$$

We also emphasize that even on totally bounded state spaces there is an interesting twist because the uniform minorization condition $\jmath(\mathcal{K})>0$ is generally not satisfied. We illustrate this assertion with the semi-circle transition on the unit interval $\mathbb{X} =]0,1[= \mathbb{Y}$ defined by

$$\mathcal{K}(x,dy) = \frac{1}{q(x)}\ \sqrt{1-(x-y)^2}\ 1_{]0,1[}(y)\ dy \tag{3.11}$$

for some normalizing constant $1/3 \leqslant q(x) \leqslant 1$. In this case, choosing $x_1 = 1/(2n) = 1-y = 1-x_2$ in (3.7) we check that

$$0 \leqslant \jmath(\mathcal{K}) \leqslant 3\ \sqrt{1-(1-1/n)^2} \stackrel{n\to\infty}{\longrightarrow} 0 \Longrightarrow \jmath(\mathcal{K}) = 0.$$

The entropic transport model (1.4) on the unit interval $]0,1[$ with reference transition (3.11) is discussed in Section 5.4. Note that in that case, the cost function is unbounded.

3.2.2. *Dobrushin contraction coefficients.* Consider functions $\varphi \in \mathcal{B}(\mathbb{X})$ and $\psi \in \mathcal{B}(\mathbb{Y})$ such that $\varphi(x)\wedge\psi(y) \geqslant 1$ for any $(x,y)\in(\mathbb{X}\times\mathbb{Y})$. The (φ,ψ)-Dobrushin coefficient $\chi_{\varphi,\psi}(\mathcal{K})$ of a Markov transition $\mathcal{K}(x,dy)$ from $\mathbb{X}$ into $\mathbb{Y}$ is defined by the norm operator

$$\chi_{\varphi,\psi}(\mathcal{K}) = \sup_{\nu_1,\nu_2\in\mathcal{M}_{1,\varphi}(S)} \frac{|||\nu_1\mathcal{K}-\nu_2\mathcal{K}|||_\psi}{|||\nu_1-\nu_2|||_\varphi}.$$

When $(\varphi,\mathbb{X}) = (\psi,\mathbb{Y})$ we may also write $\chi_\varphi(\mathcal{K})$ instead of $\chi_{\varphi,\varphi}(\mathcal{K})$. Given a Markov transition $\mathcal{K}_0(x,dy)$ from $\mathbb{X}$ into $\mathbb{Y}$ and a Markov transition $\mathcal{K}_1(x,dy)$ from $\mathbb{Y}$ into $\mathbb{X}$ we have

$$\chi_\varphi(\mathcal{K}_0\mathcal{K}_1) = \chi_{\varphi,\varphi}(\mathcal{K}_0\mathcal{K}_1) \leqslant \chi_{\varphi,\psi}(\mathcal{K}_0)\ \chi_{\psi,\varphi}(\mathcal{K}_1). \tag{3.12}$$

The terminology φ-Dobrushin coefficient comes from the fact that we recover the standard Dobrushin contraction coefficient $\chi(\mathcal{K})$ by choosing constant functions. Indeed, if $\varphi = 1 = \psi$, then

$$\chi_{1,1}(\mathcal{K}) = \sup_{\nu_1,\nu_2\in\mathcal{M}_1(\mathbb{X})} \frac{\|\nu_1\mathcal{K}-\nu_2\mathcal{K}\|_{\mathsf{tv}}}{\|\nu_1-\nu_2\|_{\mathsf{tv}}} =: \chi(\mathcal{K}) = 1-\varepsilon(\mathcal{K}), \tag{3.13}$$

with the parameter

$$\varepsilon(\mathcal{K}) := 1 - \sup_{(x_1,x_2)\in\mathbb{X}^2} \|(\delta_{x_1} - \delta_{x_2})\mathcal{K}\|_{\mathsf{tv}}. \tag{3.14}$$

Whenever $\delta_{x_1}\mathcal{K} \ll \delta_{x_2}\mathcal{K}$ for any $(x_1, x_2) \in \mathbb{X}^2$ using the estimate (3.9) as well as (3.3) we readily find that

$$\varepsilon(\mathcal{K}) \geqslant \jmath(\mathcal{K}) \geqslant \hbar(\mathcal{K}) \quad \text{and} \quad \chi(\mathcal{K}) \leqslant 1 - \jmath(\mathcal{K}) \leqslant 1 - \hbar(\mathcal{K}). \tag{3.15}$$

3.2.3. *Entropy contraction coefficient.* Given a convex function Φ satisfying (1.12), the Dobrushin Φ-contraction coefficient $\chi_\Phi(\mathcal{K})$ of some Markov transition $\mathcal{K}(x, dy)$ from $\mathbb{X}$ into $\mathbb{Y}$ is defined by

$$\chi_\Phi(\mathcal{K}) := \sup_{\nu_1,\nu_2\in\mathcal{M}_1(\mathbb{X})} \frac{D_\Phi(\nu_1\mathcal{K}, \nu_2\mathcal{K})}{D_\Phi(\nu_1, \nu_2)}, \tag{3.16}$$

where the supremum is taken over all probability measures $\nu_1, \nu_2 \in \mathcal{M}_1(S)$ such that $D_\Phi(\nu_1, \nu_2) > 0$. Here, again, the terminology Φ-Dobrushin coefficient comes from the fact that choosing $\Phi(u, v) = \Phi_0(u, v) = \frac{1}{2}|u - v|$ we recover the standard Dobrushin contraction coefficient, i.e., $\chi_{\Phi_0}(\mathcal{K}) = \chi(\mathcal{K})$. For any convex function Φ satisfying (1.12) we recall that

$$\chi_\Phi(\mathcal{K}) \leqslant \chi(\mathcal{K}). \tag{3.17}$$

The estimate (3.17) ensures that the Dobrushin coefficient is a universal contraction coefficient (see [24] for finite state spaces and [43] for general measurable spaces). For further details on the Dobrushin contraction coefficient we refer the reader to the article [48], see also [40, 43] and Chapter 4 in [38].

4. Sinkhorn contraction properties

This section investigates the contraction and stability properties of Sinkhorn iterations and their associated Gibbs-loop semigroups. We first recall the strong-mixing regime, in which uniform positivity of the reference kernel yields Hilbert-metric contractions and exponential convergence. We then move beyond this setting by introducing Lyapunov drift and local minorization conditions tailored to the forward-backward Sinkhorn transitions. These conditions allow us to establish weighted-norm contraction inequalities, from which we derive exponential convergence rates in total variation, weighted norms, Wasserstein distances, and relative entropy. The results provide a unified framework for analyzing Sinkhorn dynamics on non-compact state spaces and for treating models that fall outside the classical strong-mixing assumptions.

4.1. **Strong mixing models.** As underlined in [58, 82] in the context of finite state space models, Sinkhorn iterations of the form (2.2) preserve the cross product Radon-Nikodym ratio in (3.8) and, hence, for any $n \geqslant 0$ we have

$$\hbar(\mathcal{S}_n) = \hbar(\mathcal{S}_0) = \hbar(\mathcal{Q}). \tag{4.1}$$

This property is crucial in the convergence analysis of Sinkhorn iterations using Hilbert-projective techniques, see for instance [15, 18, 46, 60]. Recall that the Hilbert contraction coefficient of a Markov transition $\mathcal{Q}$ can be explicitly computed in terms of $\hbar(\mathcal{Q})$ (see Eq. (3.10)), hence the exponential convergence of Sinkhorn iterations w.r.t. the Hilbert metric takes place if, and only if, $\hbar(\mathcal{Q}) > 0$. An elementary proof of this property is presented in Proposition B.1 (in Appendix B). From (3.9), let us observe that

$$\jmath(\mathcal{S}_n) \wedge \jmath(\mathcal{Q}) \geqslant \hbar(\mathcal{S}_n) = \hbar(\mathcal{Q}) \geqslant \jmath(\mathcal{Q})^2.$$

Thus, for any given convex function Φ satisfying (1.12), we can use (3.10) and (3.17) to show that

$$\hbar(\mathcal{Q}) > 0 \quad \text{if and only if} \quad \jmath(\mathcal{Q}) > 0,$$